

# Applied Statistical Modeling 2

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# Regression models for constrained, censored and truncated

**Censoring** generally occurs when we can observe the true value of a dependent variable of interest  $Y$ , only in a specific range of observations, and values below or above a certain range are "clustered" on a single value

- ▶  $y = y^*$  for all  $y < y^*$  *right-censored*, or
- ▶  $y = y^*$  for all  $y > y^*$  *left-censored*

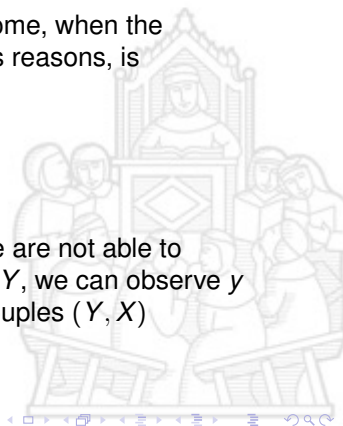


# Regression models for constrained, censored and truncated

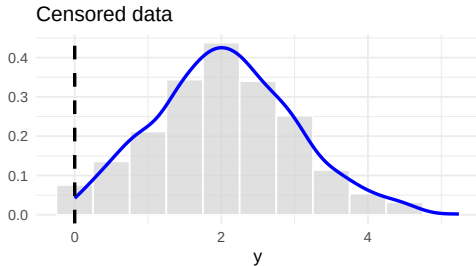
An example is the evaluation of household income, when the collection of income information, for exogenous reasons, is something like:

- ▶ household income  $> 50,000$  Euro, or
- ▶ household income  $\leq 15,000$  Euro
- ▶ ...

In any case, when censoring occurs, even if we are not able to observe the complete distribution of values for  $Y$ , we can observe  $y$  for all the  $n$  observations, so we have all the couples  $(Y, X)$

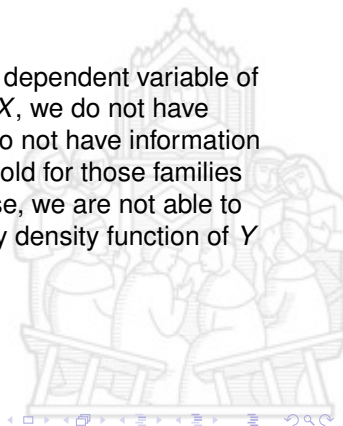


# Regression models for constrained, censored and truncated



# Regression models for constrained, censored and truncated

**Truncation** occurs when, for a given censored dependent variable of interest  $Y$ , and a set of independent variables  $X$ , we do not have information on all the couples  $(Y, X)$ , i.e., we do not have information about the type of job of the head of the household for those families whose income is below 15,000 Euro. In this case, we are not able to have complete information about the probability density function of  $Y$ .



# Tobit regression

Let's consider the censoring problem when we have values of the dependent variable of interest  $Y^*$  clustered at the minimum or maximum value, and suppose being interested in using a regression model to assess the effect of some explanatory variables  $X$ , if we think about a linear regression, we can use the model

$$y_i^* = \beta x_i + \epsilon_i$$

where  $y_i^*$  and  $x_i$  represent the values of the dependent and the explanatory variable, respectively, and  $\epsilon_i$  is the error term

# Tobit regression

Because  $Y^*$  is censored the observed variable  $Y$  could be defined as

$$y_i = \begin{cases} c & \text{if } y^* \leq c \\ y_i^* & \text{if } y^* > c \end{cases}$$

where  $c$  is the minimum value observed for  $y_i^*$

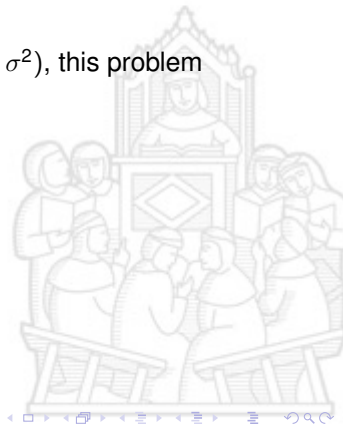
Similar reasoning for  $c$  being the maximum value observed



# Tobit regression

If  $Y^*$  is non-negative and we assume  $\epsilon_i \sim N(0, \sigma^2)$ , this problem could be solved with a Tobit regression

$$y_i^* = \beta x_i + \epsilon_i$$



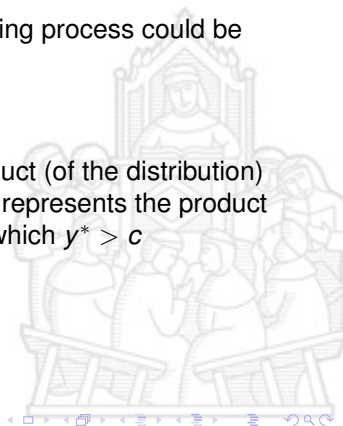


# Tobit regression

The likelihood function according to the censoring process could be expressed as

$$L = \prod_0 F_i(c_i) \prod_1 f_i(y_i)$$

with  $i = 1, \dots, n$  and  $\prod_0 F_i$  representing the product (of the distribution) over observations for which  $y^* \leq c$ , while  $\prod_1 f_i$  represents the product (of the density function) over observations for which  $y^* > c$

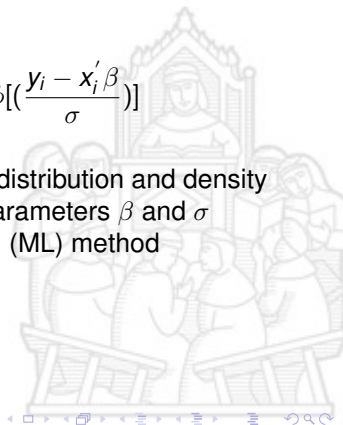


# Tobit regression

That is

$$L(\beta, \sigma) = \prod_0 [1 - \Phi(\frac{c - x_i' \beta}{\sigma})] \prod_1 \frac{1}{\sigma} \phi[(\frac{y_i - x_i' \beta}{\sigma})]$$

with  $i = 1, \dots, n$ ,  $\Phi$  and  $\phi$  being respectively the distribution and density function of the standard normal variable and parameters  $\beta$  and  $\sigma$  could be estimated by the Maximum Likelihood (ML) method



# Tobit regression

In the standard Tobit model when

$$y_i = \begin{cases} 0 & \text{if } y^* \leq 0 \\ y_i^* & \text{if } y^* > 0 \end{cases}$$

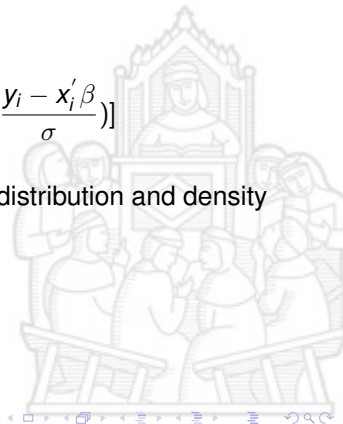


# Tobit regression

The likelihood function is given by

$$L(\beta, \sigma) = \prod_0 [1 - \Phi(\frac{x_i' \beta}{\sigma})] \prod_1 \frac{1}{\sigma} \phi[(\frac{y_i - x_i' \beta}{\sigma})]$$

with  $i = 1, \dots, n$ ,  $\Phi$  and  $\phi$  being respectively the distribution and density function of the standard normal variable

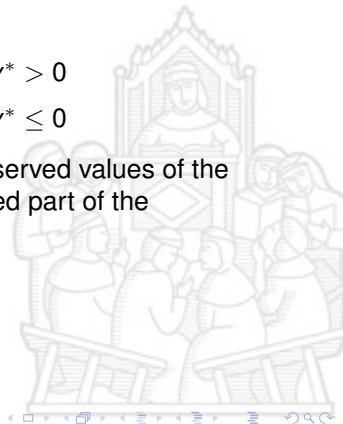


# Tobit regression

Or similarly

$$L_i(\beta, \sigma) = \begin{cases} \frac{1}{\sigma} \phi \frac{y_i - x_i' \beta}{\sigma} & \text{if } y^* > 0 \\ 1 - \frac{1}{\sigma} \Phi \frac{x_i' \beta}{\sigma} & \text{if } y^* \leq 0 \end{cases}$$

that is a combination of a linear part for the observed values of the latent variable and a Probit part for the truncated part of the distribution (of the latent variable)

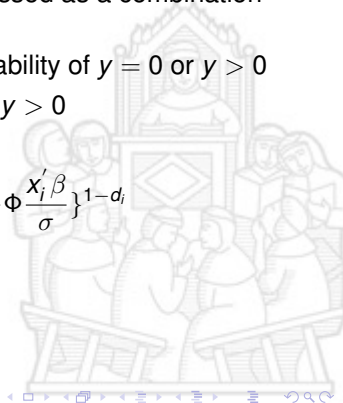


# Tobit regression

The standard Tobit model, could then be expressed as a combination of a

- ▶ a Probit model which determines the probability of  $y = 0$  or  $y > 0$
- ▶ a truncated (normal) regression model for  $y > 0$

$$L_i(\beta, \sigma) = \left\{ \frac{1}{\sigma} \phi \frac{y_i - x_i' \beta}{\sigma} \right\}^{d_i} * \left\{ 1 - \frac{1}{\sigma} \Phi \frac{x_i' \beta}{\sigma} \right\}^{1-d_i}$$



# Tobit regression

In the model proposed, we are essentially interested in estimating two different kinds of expectations

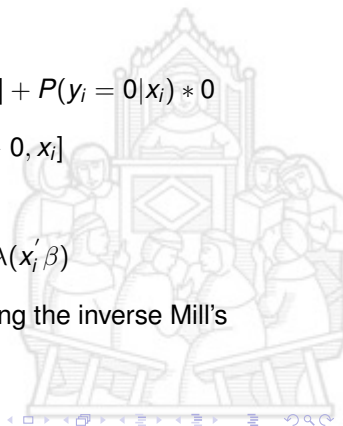
- ▶ Unconditional expectation

$$\begin{aligned} E[y_i|x_i] &= P(y_i > 0|x_i) * E[y_i|y_i > 0, x_i] + P(y_i = 0|x_i) * 0 \\ &= P(y_i > 0|x_i) * E[y_i|y_i > 0, x_i] \end{aligned}$$

- ▶ Conditional expectation

$$E[y_i|y_i < 0, x_i] = x_i' \beta + \sigma \lambda(x_i' \beta)$$

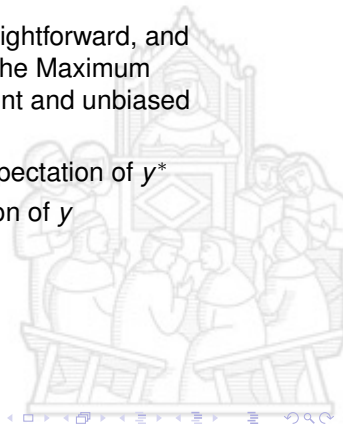
as  $E[y^*|y > c] = \mu^* + \sigma \lambda(c - \mu^*)/\sigma$  with  $\lambda$  being the inverse Mill's ratio



# Tobit regression

Parameters estimation in this setting is not straightforward, and several approaches have been proposed with the Maximum Likelihood (ML) estimation resulting in consistent and unbiased estimation of

- ▶  $\beta_k$  estimate the linear effect of  $x$  on the expectation of  $y^*$
- ▶  $\beta$  estimate the effect of  $x$  on the expectation of  $y$





# Tobit regression

If we are interested in estimating the effect of  $x$  on the observed variable  $y$  we need to assess the marginal effect

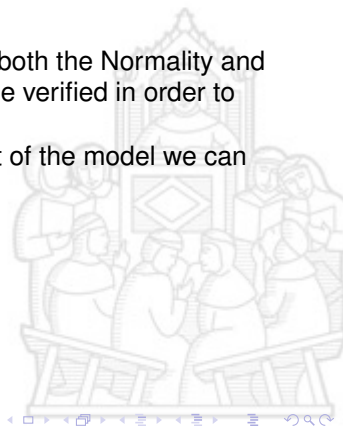
$$\Phi \frac{(x_i' \beta)}{\sigma} \beta_k$$



# Tobit regression

The consistency of the ML estimates relies on both the Normality and homoskedasticity assumptions, which should be verified in order to ensure the validity of the model

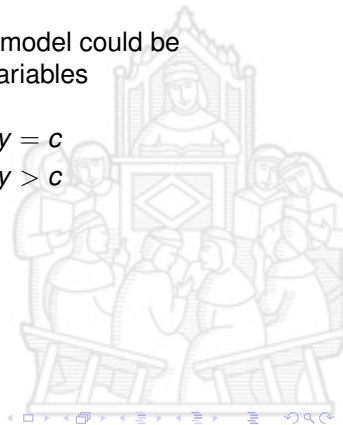
Also for the Tobit, to assess the goodness-of-fit of the model we can rely on a pseudo  $R^2$



# Tobit regression

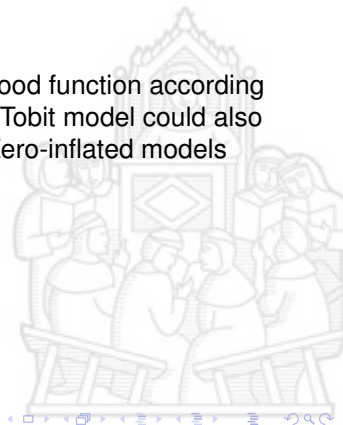
Moreover, the reasoning for the standard Tobit model could be extended to more general cases of censored variables

$$f(x; \mu, \sigma, c) = \begin{cases} \Phi\left(\frac{c-\mu}{\sigma}\right) & \text{if } y = c \\ \frac{1}{\sigma}\phi\left(\frac{y-\mu}{\sigma}\right) & \text{if } y > c \end{cases}$$



# Zero Inflated models

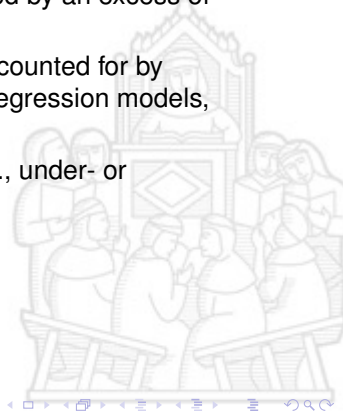
The general concept of "partitioning" the likelihood function according to the censoring process we have seen for the Tobit model could also be useful to understand two-part models and Zero-inflated models



# Zero Inflated models

In some applications, data may be characterized by an excess of zeroes, which results in

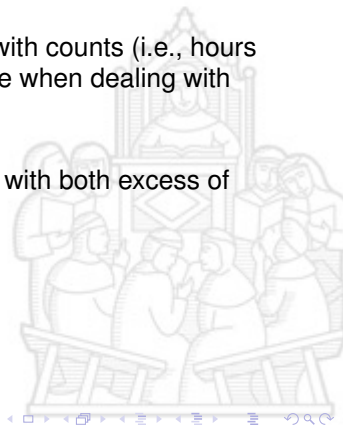
- ▶ excess of variability which could not be accounted for by common random variables and standard regression models, **overdispersion**
- ▶ biased estimates of model coefficients (i.e., under- or over-estimate of the effect)
- ▶ poor model fit



# Zero Inflated models

Zero-inflation commonly occurs when dealing with counts (i.e., hours of work in a population), but they may also arise when dealing with proportions

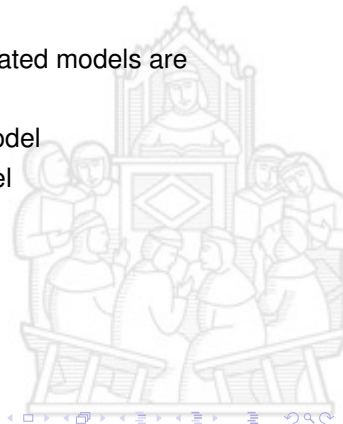
In the latter case, we may also have a problem with both excess of zeroes and of ones



# Zero Inflated models

Accordingly, the most commonly used zero-inflated models are

- ▶ Zero-Inflated Poisson (ZIP) model
- ▶ Zero-Inflated Negative Binomial (ZINB) model
- ▶ Zero(and/or One) Inflated Beta (ZIB) model



# Zero Inflated models

For each of the zero-inflated models, an observed variable  $Y$  is assumed to be generated by two underlying processes

- ▶ a Bernoulli process determining whether  $y_i$  is a **structural zero**, or a count (including some zeroes)
- ▶ a Poisson, Negative Binomial or Beta process generating counts (including some zeroes)

